

Regression 😊

Predictive analytics

Objectives

- What is regression (1) Predictor (SS) ✓
- how to assess the fit of regression model
 - SST ✓
 - SSE ✓
 - SSR ✓
 - R^2
 - F-statistic } Python will provide
- Interpret a regression model

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- A **regression** is a way of **predicting** a value for a given variable given a set of variables.

→ hypothetical model based on relationships

→ **Linear Regression**: linear model; the relationship can be described with linear equation

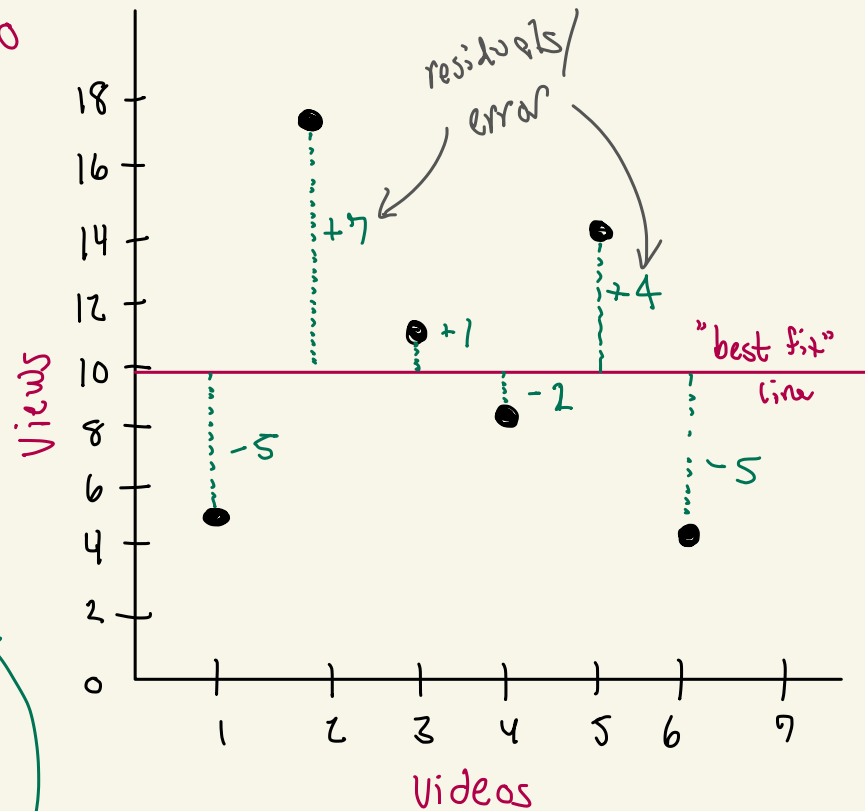
$$y = mx + b$$

ELIS: We want a line to predict values.

Example: Predicting Values

x	y
Video	Views
1	5
2	17
3	11
4	8
5	14
6	5
7	?

Mean prediction: $\bar{y} = 10$



Goodness of fit (Residuals)

- Residuals should sum to 0

Video	Views	Residual	Residual ²
1	5	-5	25
2	17	7	49
3	11	1	1
4	8	-2	4
5	14	4	16
6	5	-5	25
7	?		

$$\Sigma = 0$$

$$\Sigma = 120$$

We square because it makes all values positive and exaggerates larger errors.

$$SST (\text{sum of squares total}) = \Sigma \text{residuals}^2 = 120$$

Who cares? ~~☆☆☆~~

→ The goal of regression is to make a line that reduces SST

→ Lower SST = better fit line

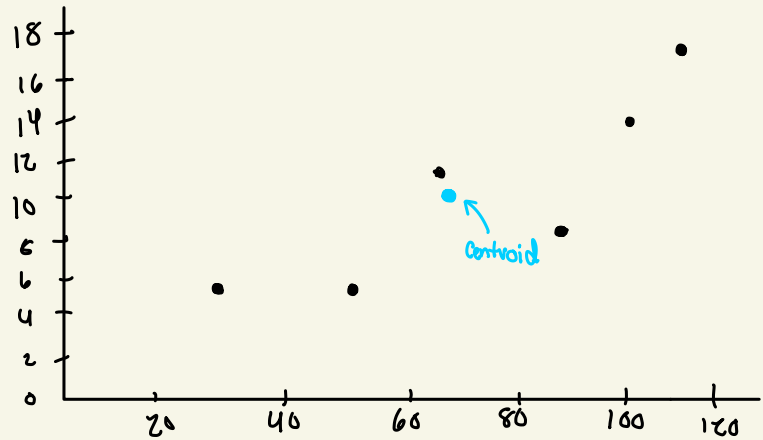
→ A good model produces a value < SST

Linear Regression for 2 Variables

- Can we provide a better prediction with the help of an additional variable?

→ This is different than mean prediction.

x	y
Stream length (min)	Views
34	5
108	17
64	11
88	8
99	14
51	5



Centroid: Calculated point which the regression line must pass through.

$$(\bar{x}, \bar{y}) = (74, 10)$$

Hypothesis: the # views is dependent on the length of the stream.

→ longer streams = more views

→ shorter streams = less views

Regression line formula

$$y_i = b_0 + b_1 x_i + \epsilon_i$$

$$b_0 = \bar{y} - b_1 \bar{x} = \text{slope}$$

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \text{intercept (y when } x=0?)$$

* for our data

$$b_0 = 10 - 0.1462(74) = -0.8188$$

$$b_1 = 0.1462$$

$$y_i = -0.8188 + 0.1462 x_i$$

Estimated Regression Values

Reality

"Machine" prediction

Video #	length (min)	Views	regression line formula	predicted Views	Error (observed - pred)	Squared error
1	34	5	$Y_i = -0.3188 + 0.1462 \cdot 34$	4.1505	0.8495	0.7217
2	108	17	$Y_i = -0.3188 + 0.1462 \cdot 108$	14.9693	2.0307	4.1237
3	64	11	$Y_i = -0.3188 + 0.1462 \cdot 64$	8.5325	2.4635	6.0688
4	88	8	$Y_i = -0.3188 + 0.1462 \cdot 88$	12.0453	-4.0453	16.3645
5	91	14	$Y_i = -0.3188 + 0.1462 \cdot 91$	13.6535	0.3465	0.1201
6	51	5	$Y_i = -0.3188 + 0.1462 \cdot 51$	6.6359	-1.6359	2.6762

• Regression squared error

→ $SSE = \sum \text{squared error}$

→ $SSE = 30.075$

$SST = 120$ $SSE = 30.075$

Sum of Squares due to regression (SSR)

$$\begin{aligned} SSR &= SST - SSE \\ &= 120 - 30.075 \\ &= 89.925 \end{aligned}$$

"We have reduced error by 89.925 using regression."

Analyzing the Model's Fit

- Coefficient of determination: r^2 or "multiple R-squared"

$$r^2 = \frac{SSR}{SST}$$

$$r^2 = \frac{89.925}{120} = 0.7493 \text{ or } \approx 75\% \quad \leftarrow \text{This is a good model} \quad \text{😊}$$

~75% of the total sum of squares (SST) can be explained by our model.

~25% is due to error

• F-Statistic / F-Ratio

$$F\text{-Statistic} = \frac{\text{improvement due to model}}{\text{difference between model + observed}}$$

★ ★ ★

★ a good model should have a large F-Statistic (↑)

★ ★ ★ ★ ★ ★

Interpreting r^2 and F-Ratio

- r^2 → how much Variance does our model explain compared to how much Variance there is to explain in the first place. $\left. \begin{array}{l} \text{higher \%} \\ = \text{better} \end{array} \right\}$
- F-Ratio → how much Variability can explain compared to how much it can't. $\left. \begin{array}{l} 1 > \text{ is better if} \\ p\text{-value is} \\ < 0.05 \end{array} \right\}$
 $\frac{\text{how good?}}{\text{how bad?}}$